

Hybrid Computing Technology

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Evolution of Computing Technology

ผศ. อัครินทร์ คุณกิตติ

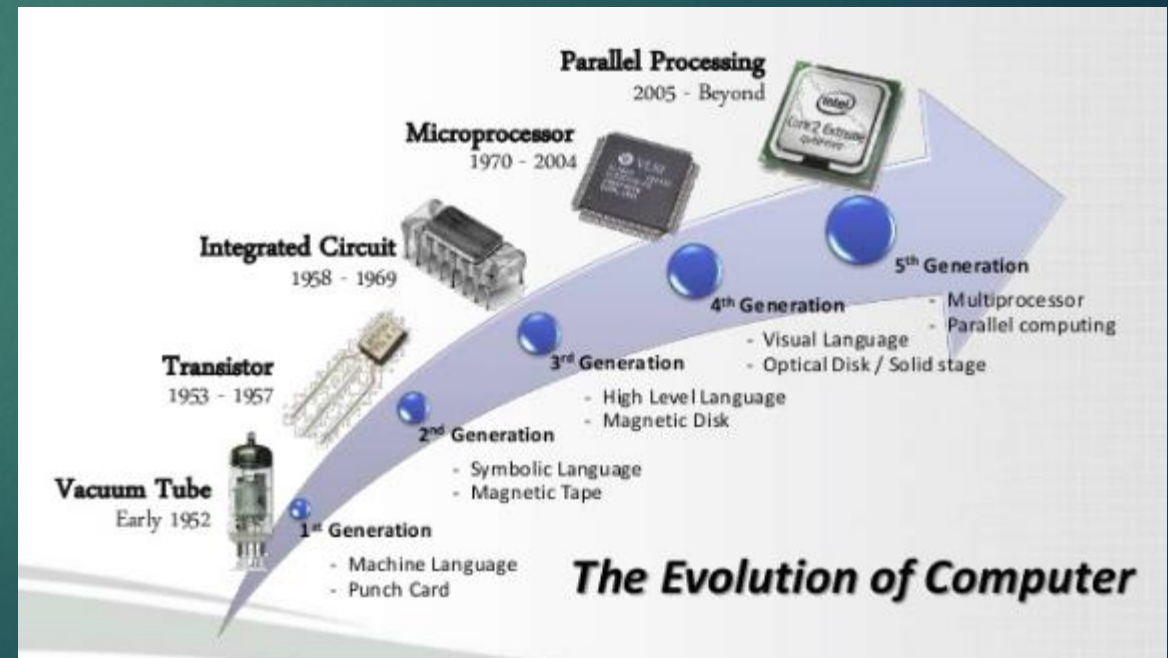
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Topics

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- ▶ Evolution of Computing Performance
- ▶ Processor Performance Evolution
- ▶ Multi-Processor / Multi-Computer
- ▶ Hardware Accelerators
- ▶ Conclusion



Computing Benchmark

- ▶ Instructions Per Second (IPS) => MIPS
- ▶ Floating point Operations Per Second (FLOPS)
 - ▶ KFLOPS, MFLOPS, GFLOPS, TFLOPS
- ▶ Single Precision (32 Bits) vs Double Precision (64 Bits)
- ▶ Other Number Formats
 - ▶ BF16 (bfloat16 - Brain Floating Point)
 - ▶ TF32 (Tensor Float 32)

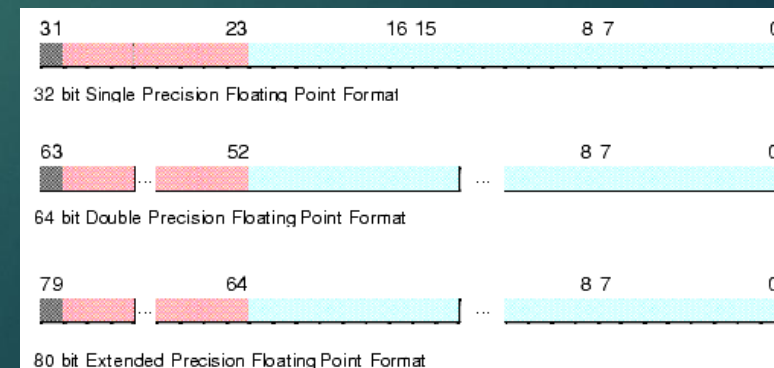
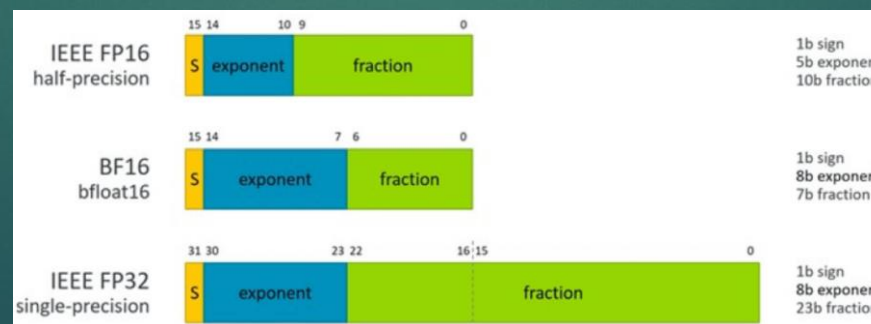
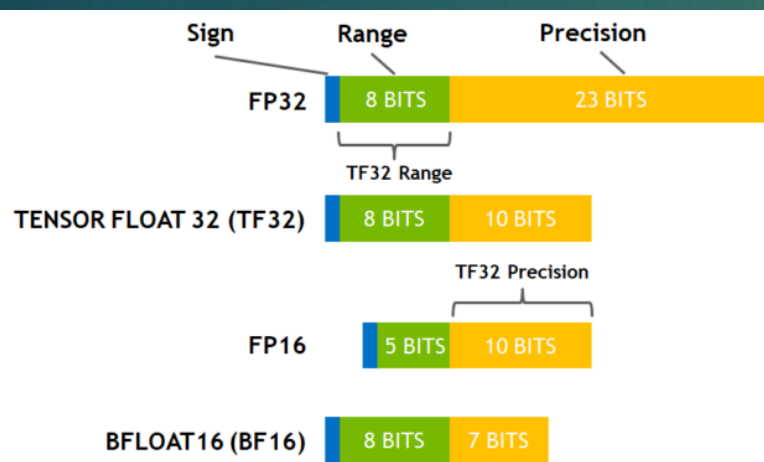
Speed	Year	Time difference
1 MFLOPS	1961	-
1 GFLOPS	1983	22 years
1 TFLOPS	1997	14 years
1 PFLOPS	2008	11 years

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Processing speed-FLOPS

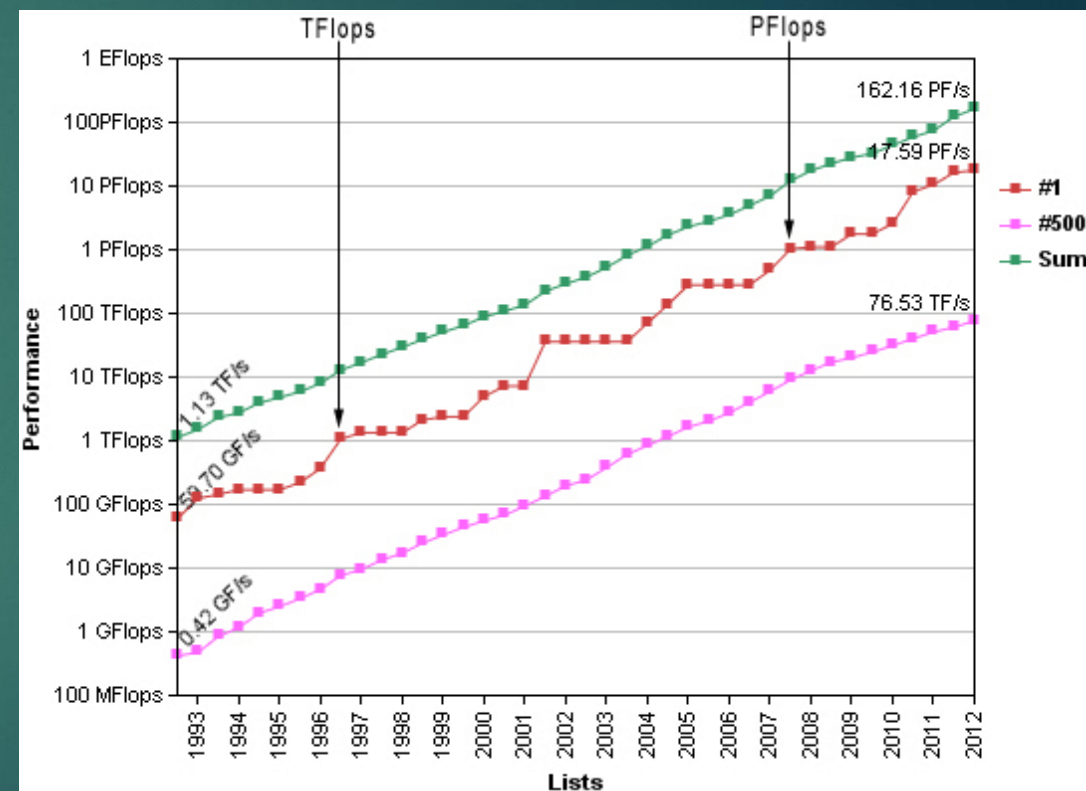
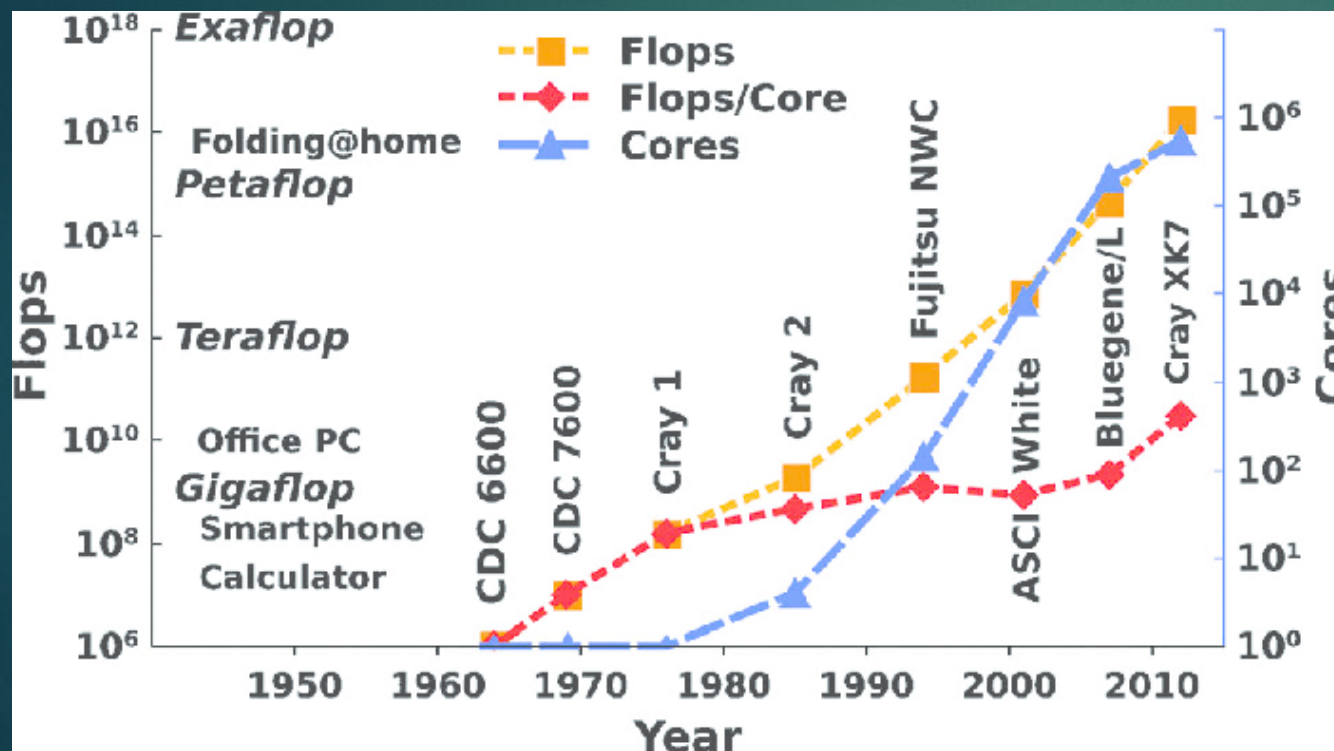
- Floating Point Operations per Second
- Measure of Computer Performance
- Best used in Scientific Calculation requiring Floating Point Calculations
- performance measure Table:

Name	Unit	Value
kiloFLOPS	kFLOPS	10^3
megaFLOPS	MFLOPS	10^6
gigaFLOPS	GFLOPS	10^9
teraFLOPS	TFLOPS	10^{12}
petaFLOPS	PFLOPS	10^{15}
exaFLOPS	EFLOPS	10^{18}
zettaFLOPS	ZFLOPS	10^{21}
yottaFLOPS	YFLOPS	10^{24}



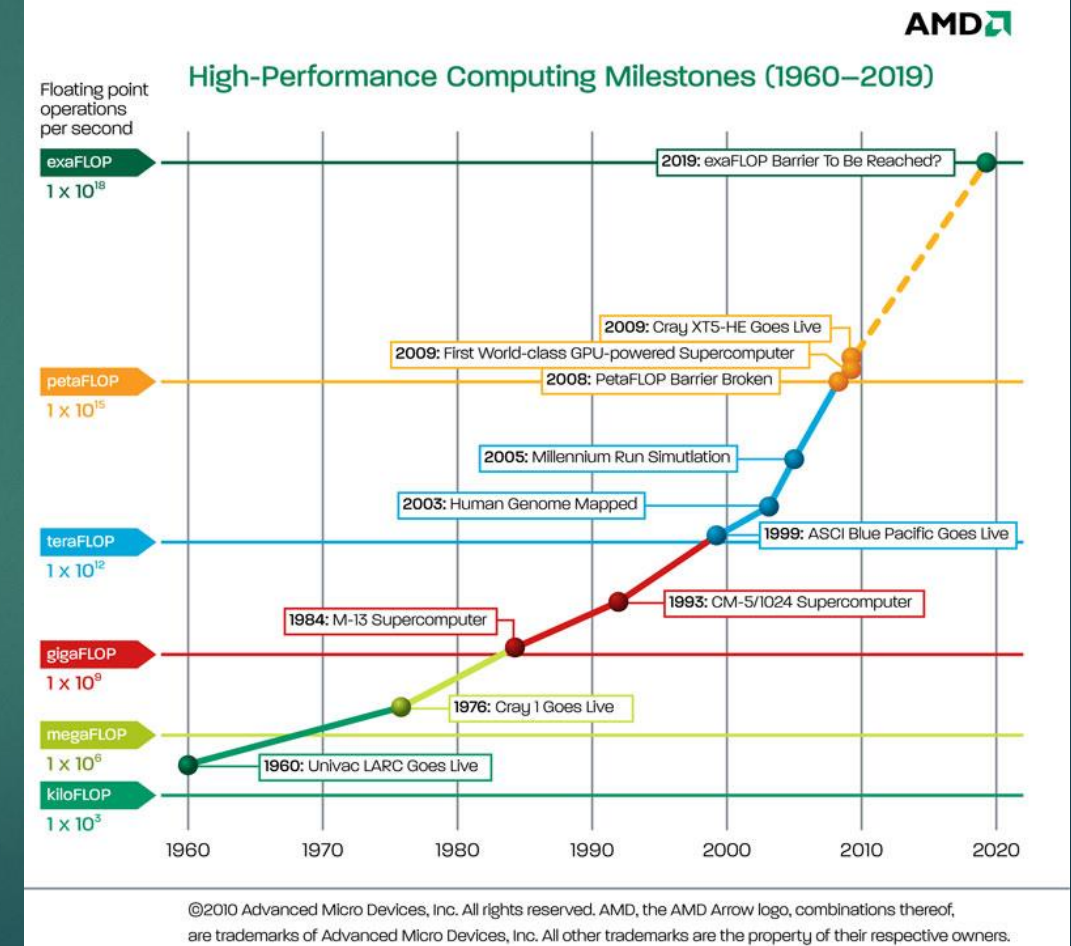
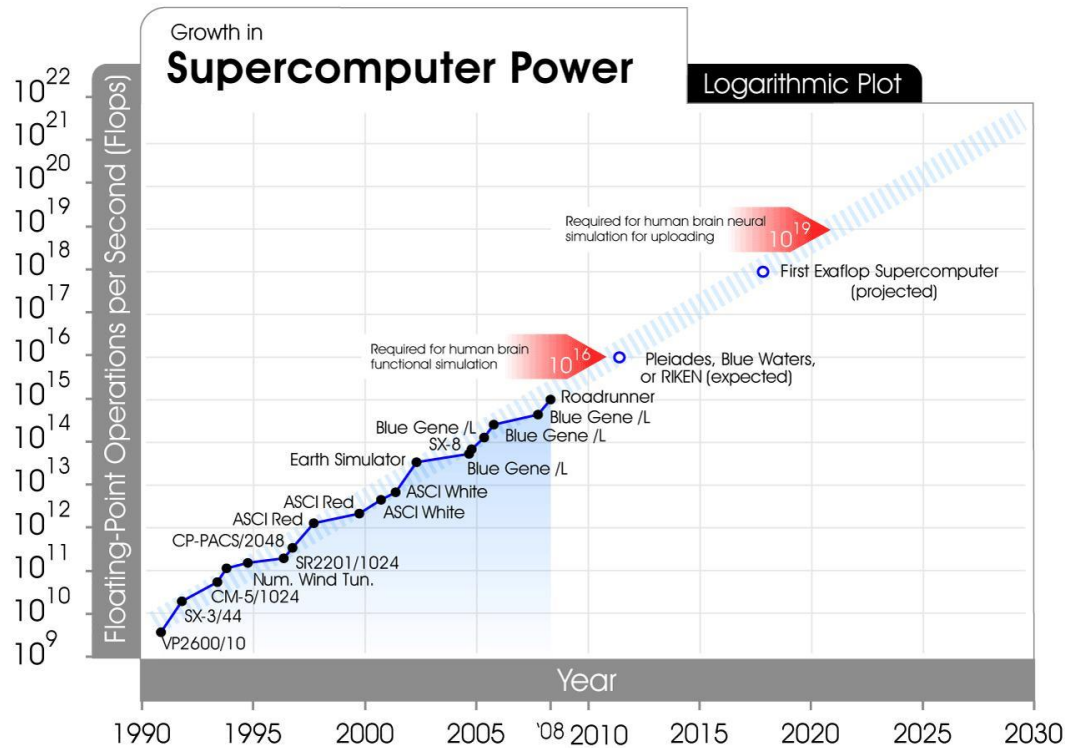
Computing Speed Evolution

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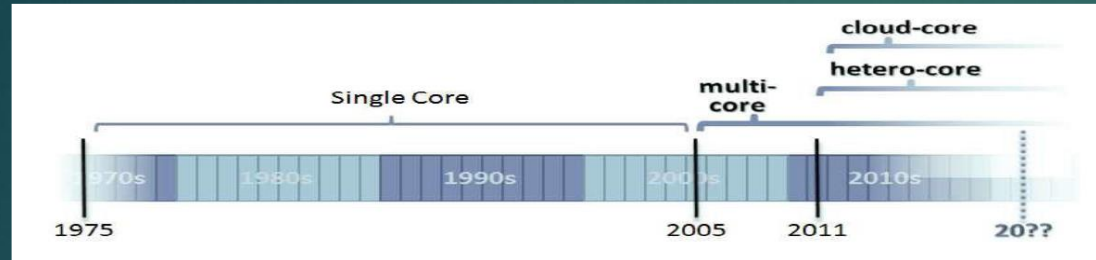
Super-Computer To High Performance Computer (HPC)

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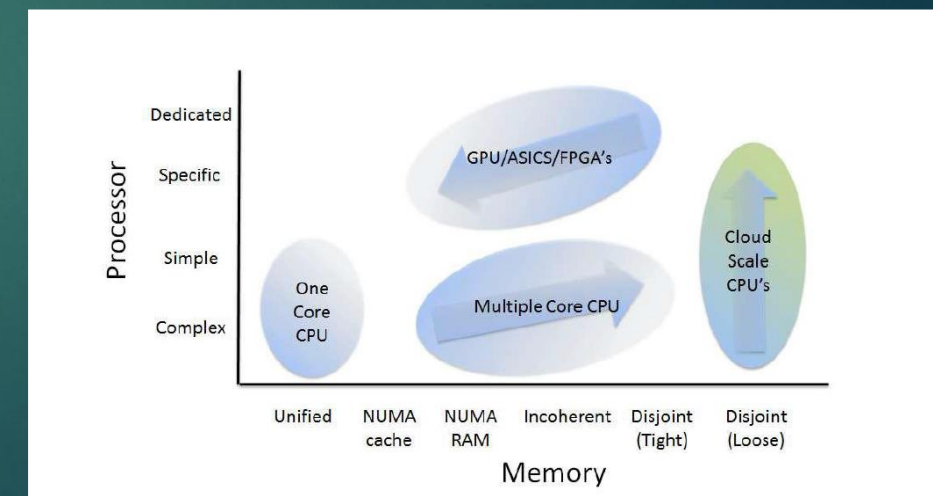
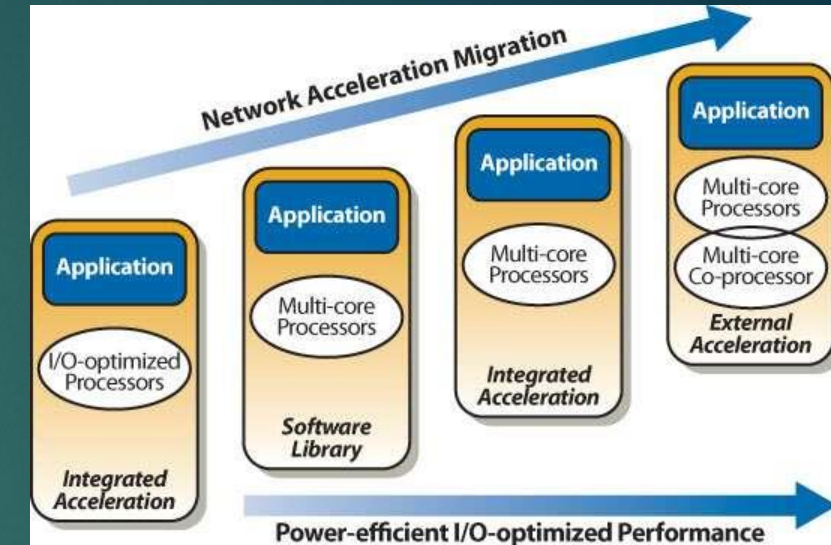


Evolution of Computing Performance

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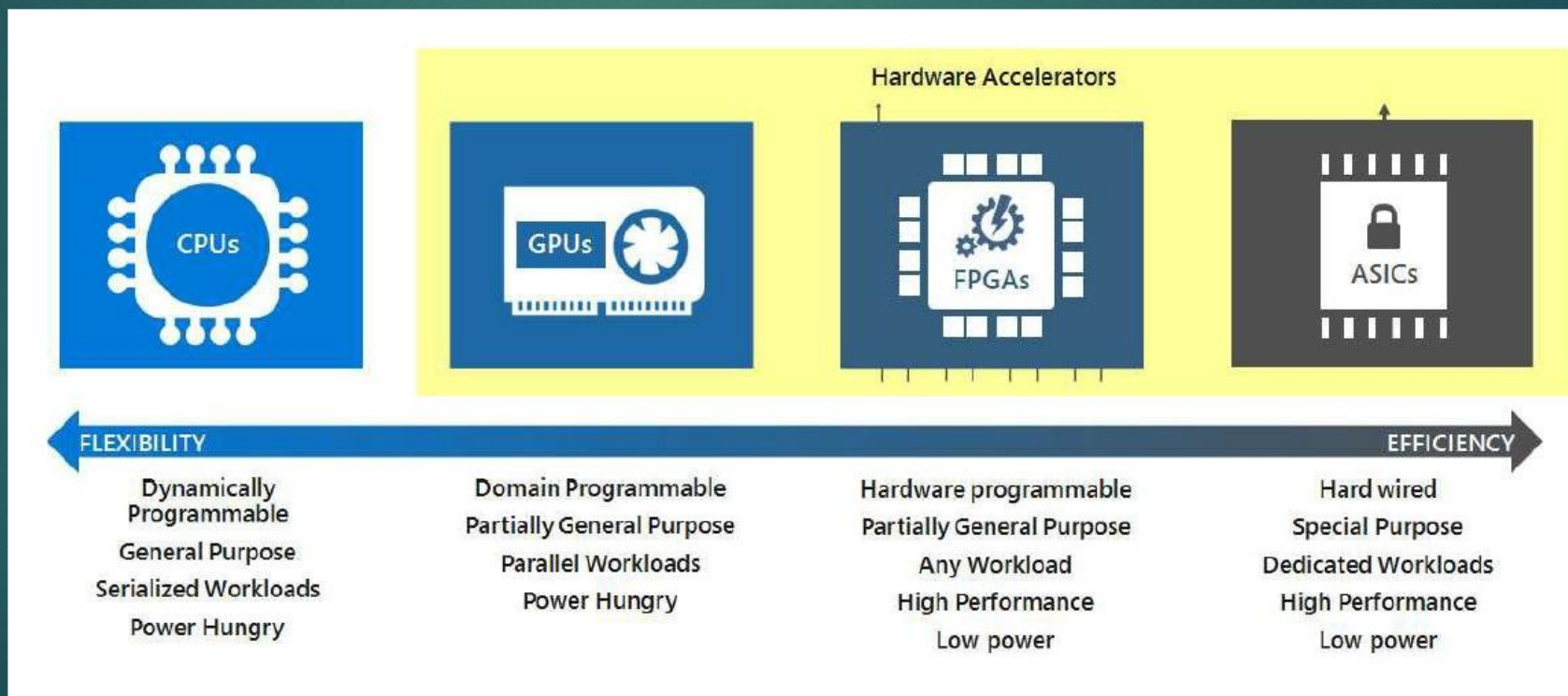
- ▶ Increase / Improve Computing Speed
- ▶ Improve both of Control and Compute Speed
- ▶ Improve Performance of Single Processor / CPU
- ▶ Improve by Multi-Processor
- ▶ Improve by Separating Tasks to Dedicated Hardware
 - ▶ Video or Graphics Processing Unit (GPU)
 - ▶ Computing Task by Accelerator



Hardware Accelerators

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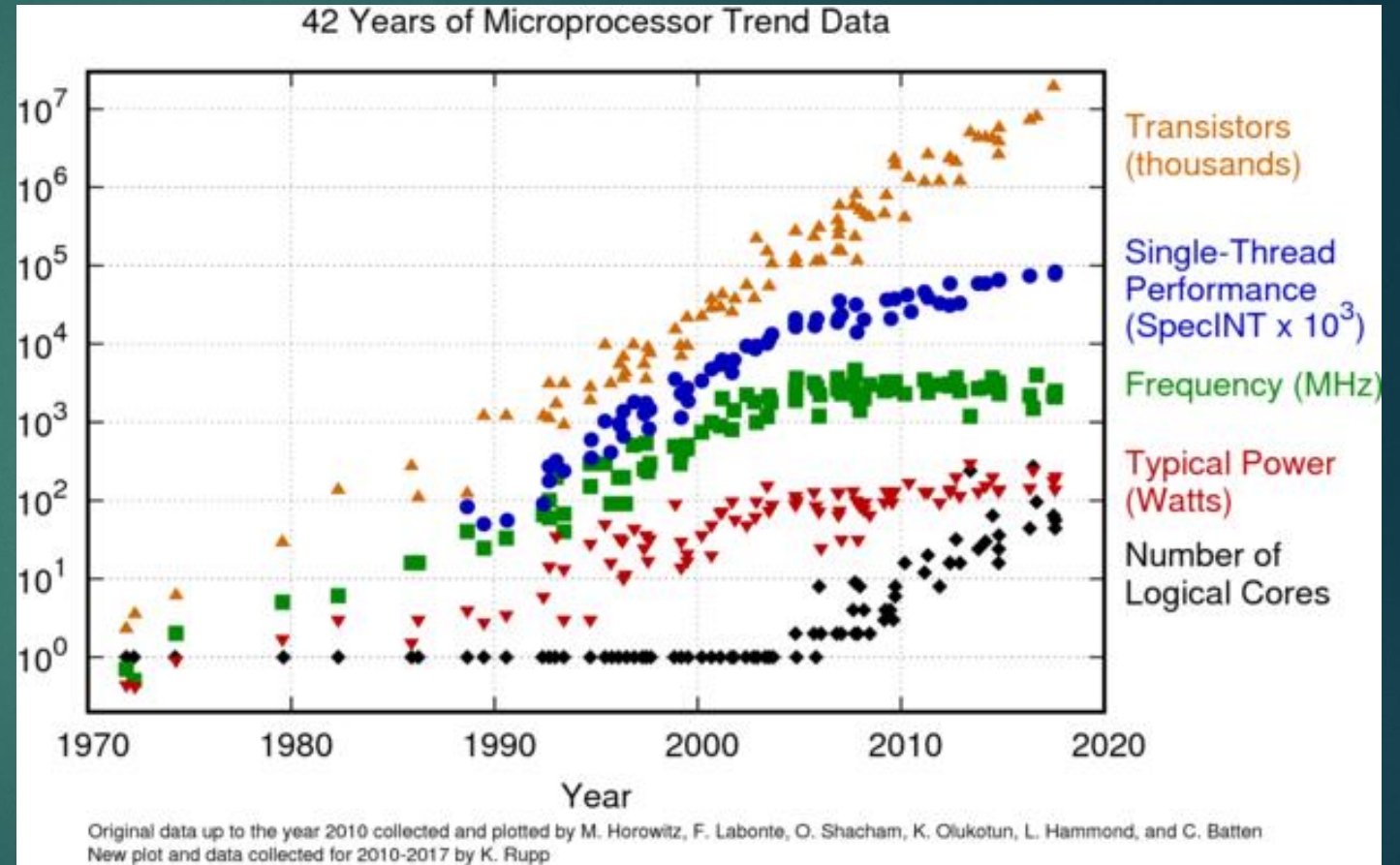
- ▶ Computing Performance Improving by Dedicated Hardware Accelerators
- ▶ Hardware Parallel Processing / Computing



Processor Performance Evolution

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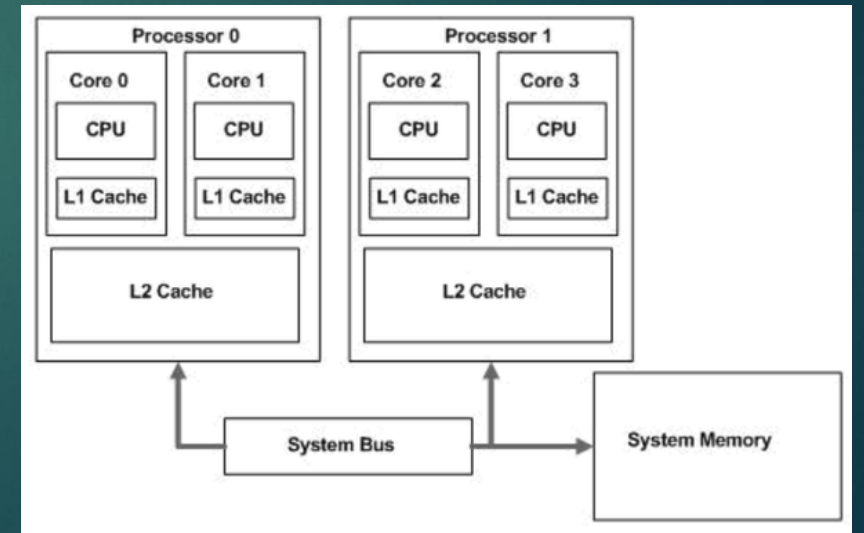
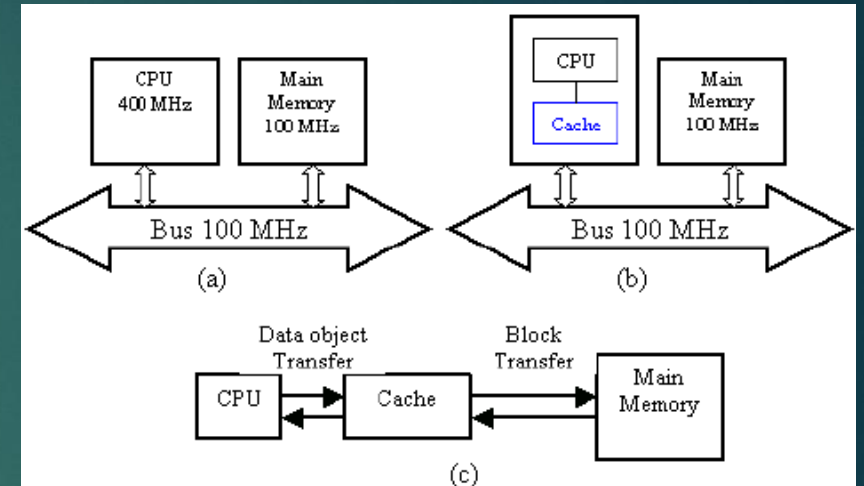
- ▶ Clock Speed
- ▶ Cache
- ▶ Pipeline
- ▶ Branch Prediction
- ▶ Co-Processor
- ▶ Multi-Core
- ▶ Specific Instructions



Processor Clock Speed & Cache

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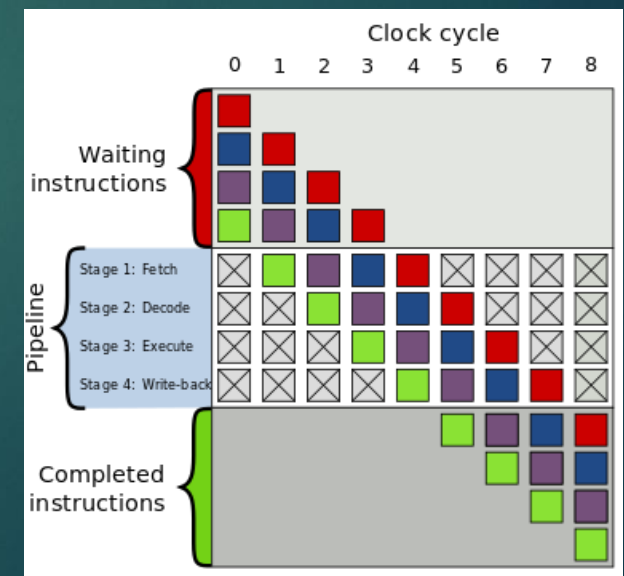
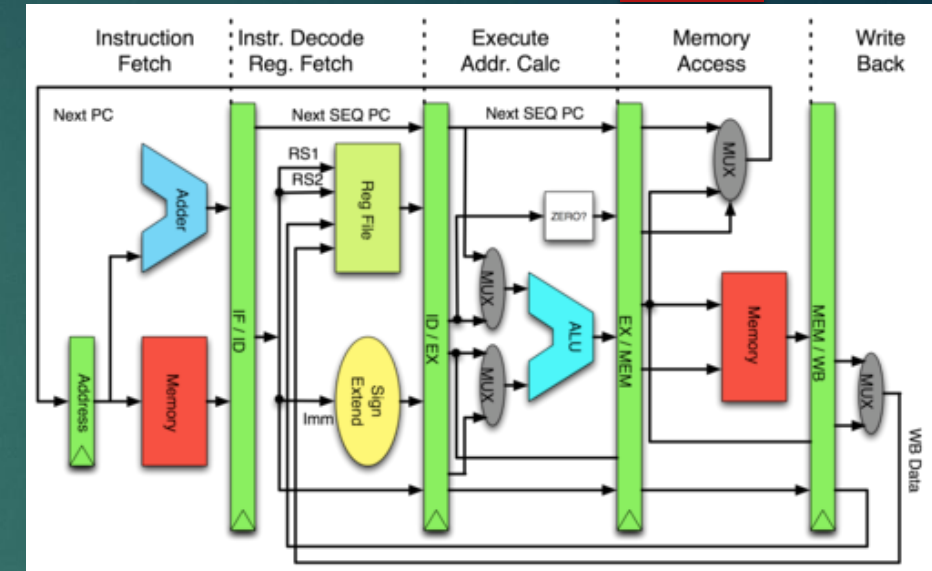
- ▶ Processor operate synchronized with clock
- ▶ Increase clock speed => Increase number of operations
- ▶ Processor speed increase, But not Memory
 - ▶ Main Memory use DRAM, slow improve
- ▶ Place higher speed storage between Processor and Main Memory, Static RAM
- ▶ Processor access to SRAM, Cache, instead of Main Memory
- ▶ Then Cache access to Main Memory if Cache missed



Instruction Pipeline

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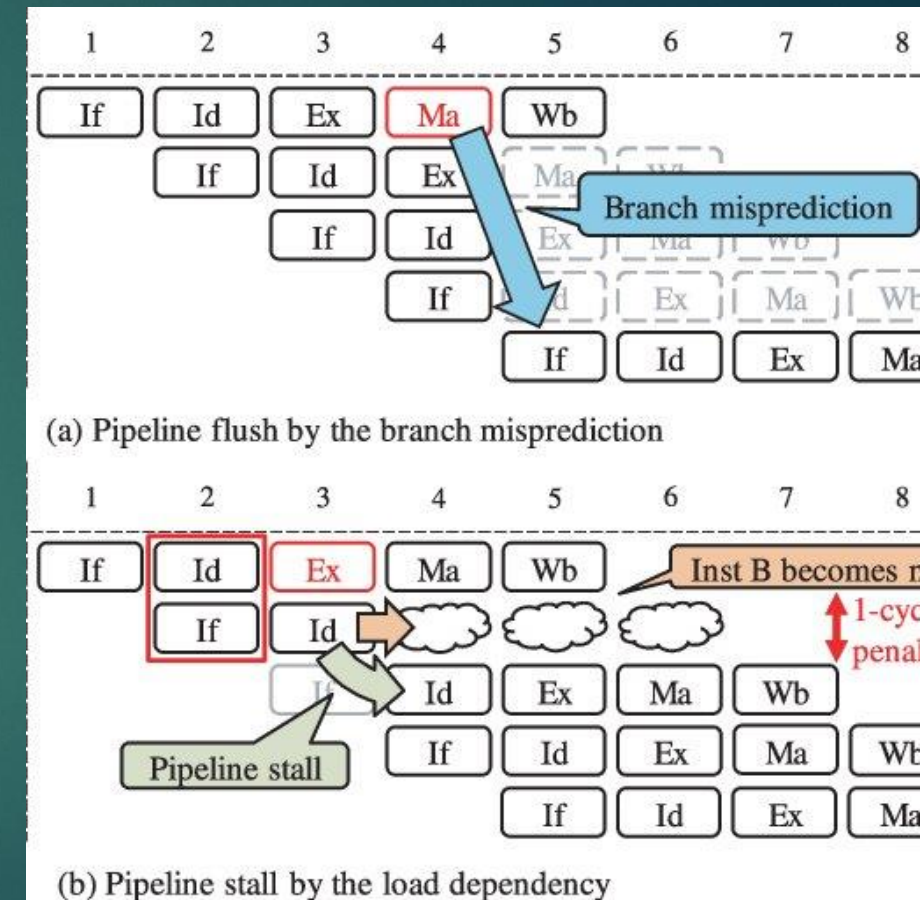
- ▶ Super Scalar
- ▶ Multiple Operations Simultaneous
 - ▶ In a same clock cycle
- ▶ Break instruction in to multiple steps
- ▶ Operate different steps of different instruction simultaneous
- ▶ Parallel operations
- ▶ Utilize resources



Branch Prediction

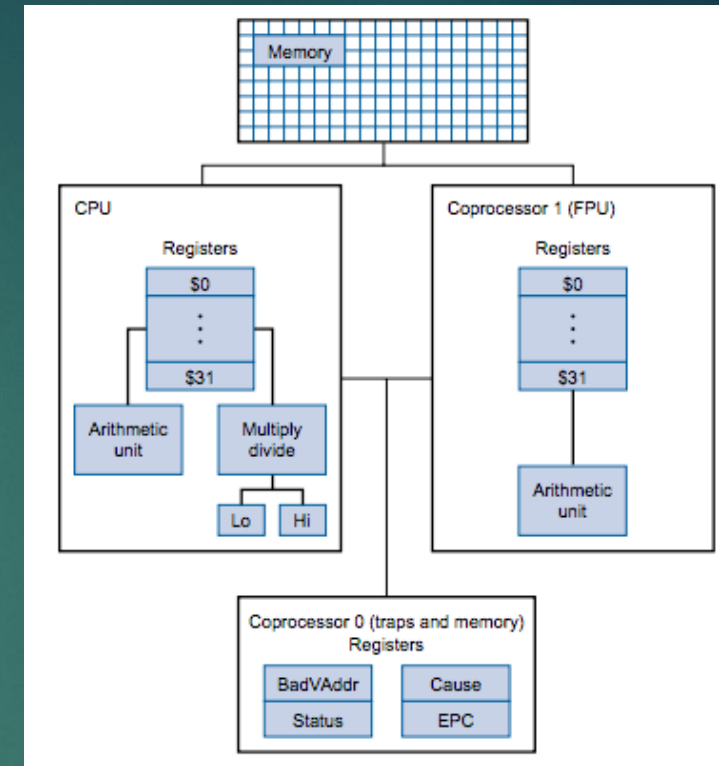
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- ▶ Effect from Pipeline Processing
- ▶ Out of Order Execution
- ▶ Instruction Flow may be changed, after execute some instruction
- ▶ Instruction Pre-Fetch/Loaded wasted
- ▶ Forward Executions from Pipeline are invalid
- ▶ Re-Load Pipeline buffers
- ▶ Performance Drop for Branch
- ▶ Many Techniques for Improve Performance



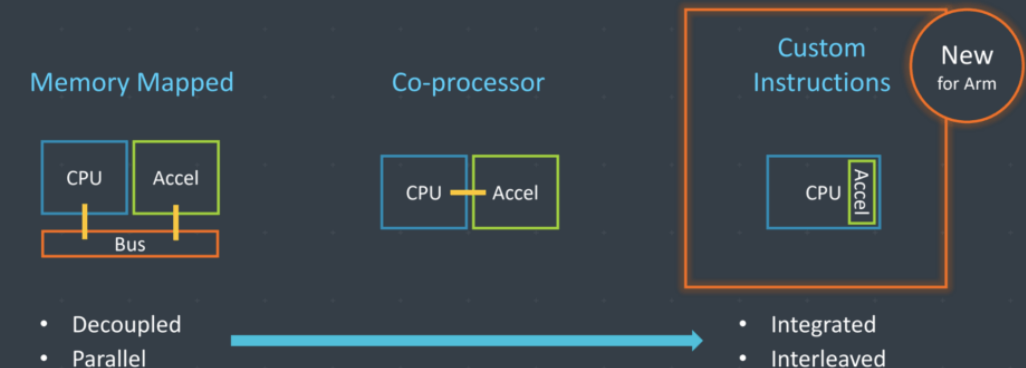
Co-Processor

- ▶ Co-operate with, or help, Processor for specific processing
 - ▶ Math Operation
 - ▶ Encrypt-Decrypt Operation
 - ▶ Digital Signal Processing (DSP)
- ▶ Specific Hardware Design
- ▶ Internal or External



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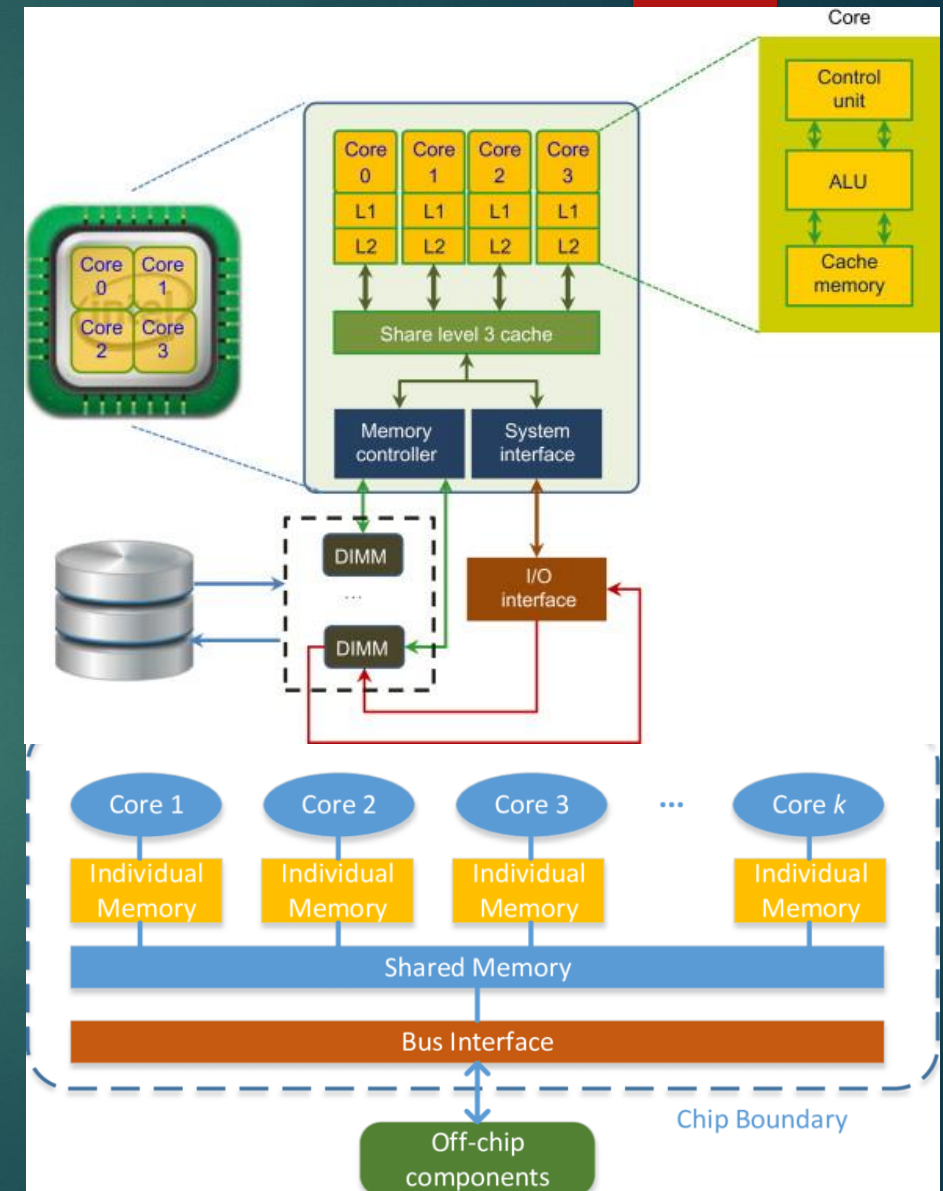
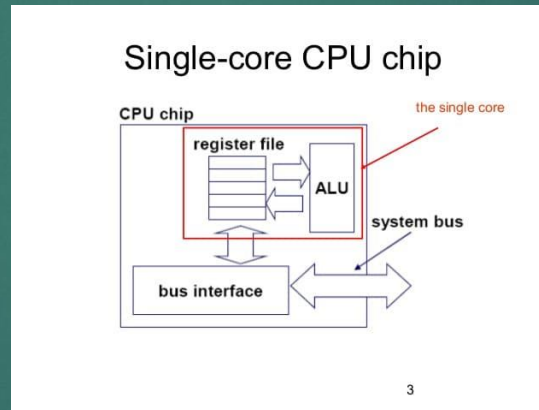
Meeting the requirements for acceleration



Multi-Core Processor

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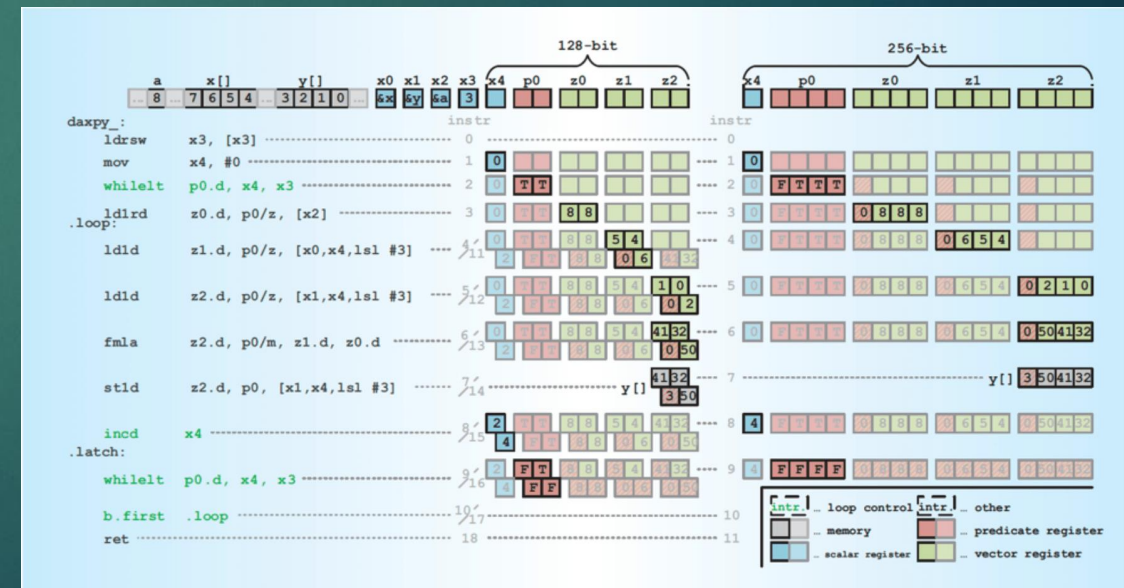
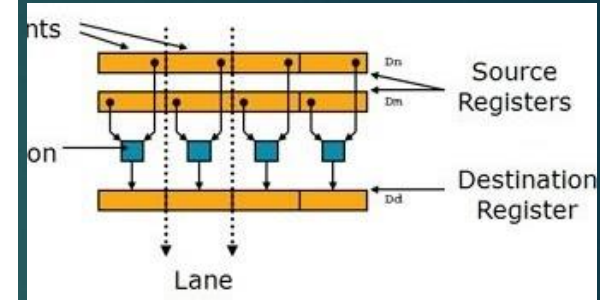
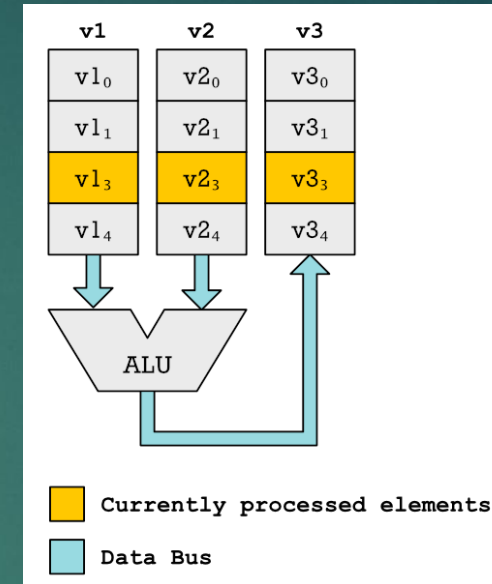
- ▶ Single Core Performance Limit
- ▶ Increase number of Processing Cores
- ▶ Logical / Virtual Core Processor
- ▶ Each Core has
 - ▶ ALU / Compute Unit
 - ▶ Registers / Cache
 - ▶ Control Unit
- ▶ Share some Resources
 - ▶ Cache / Memory
 - ▶ Bus / Control Interface



Specific Instructions

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- ▶ Create Specific Instructions (Set)
- ▶ Array / Vector Processing
- ▶ Specific Hardware Circuits for Instruction
 - ▶ ALU / Compute Unit
 - ▶ Control

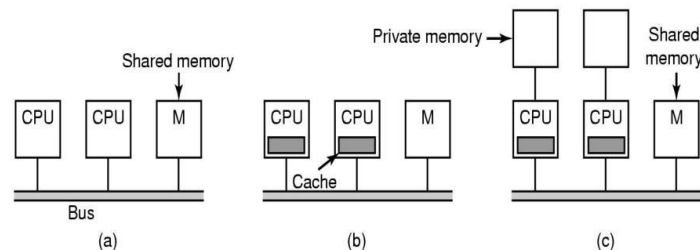


Multi-Processor

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- ▶ Multiple Processors in a System
- ▶ Each Processor may be multicore
- ▶ Parallel Processing

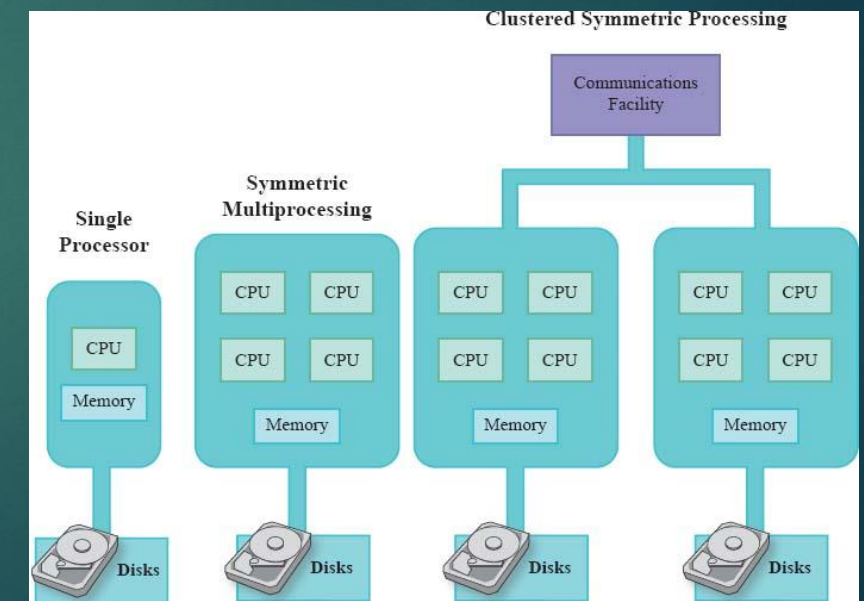
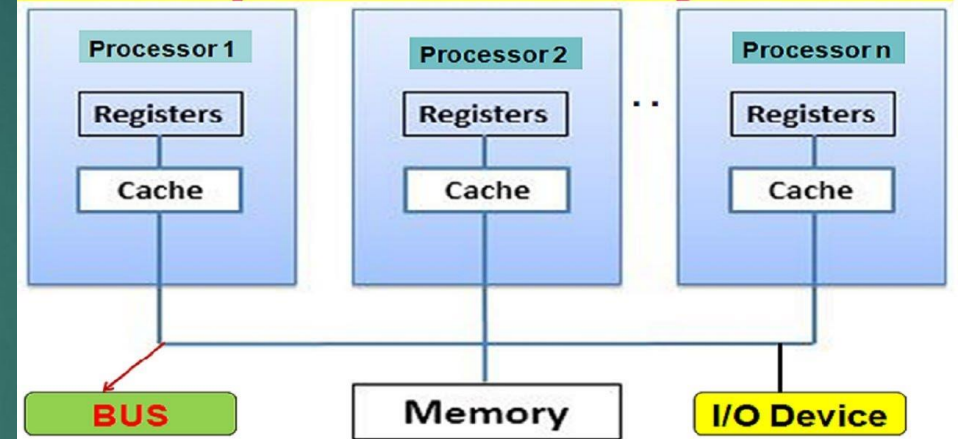
Multiprocessor Hardware (1)



Bus-based multiprocessors

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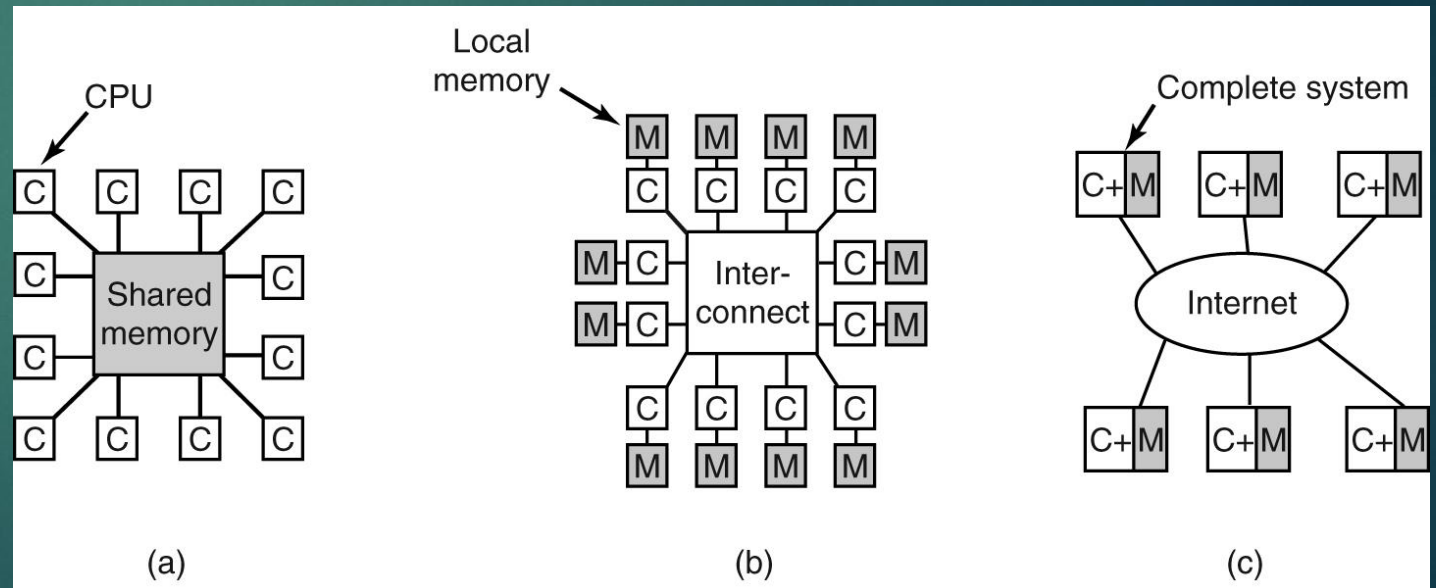
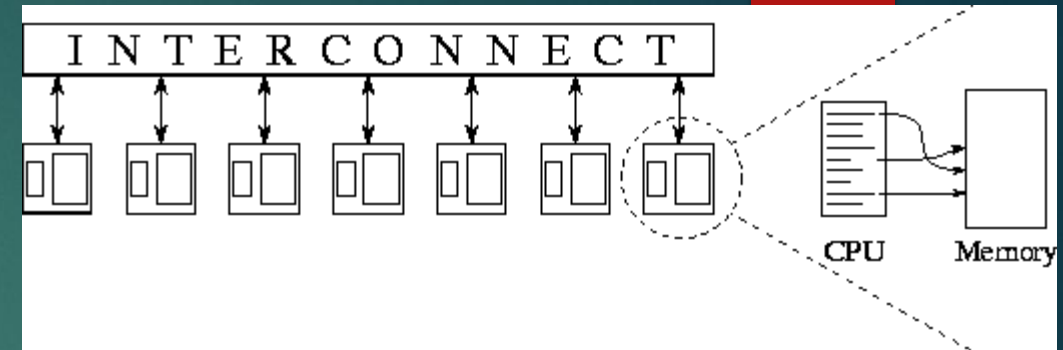
Multiprocessors Systems



Multi-Computer

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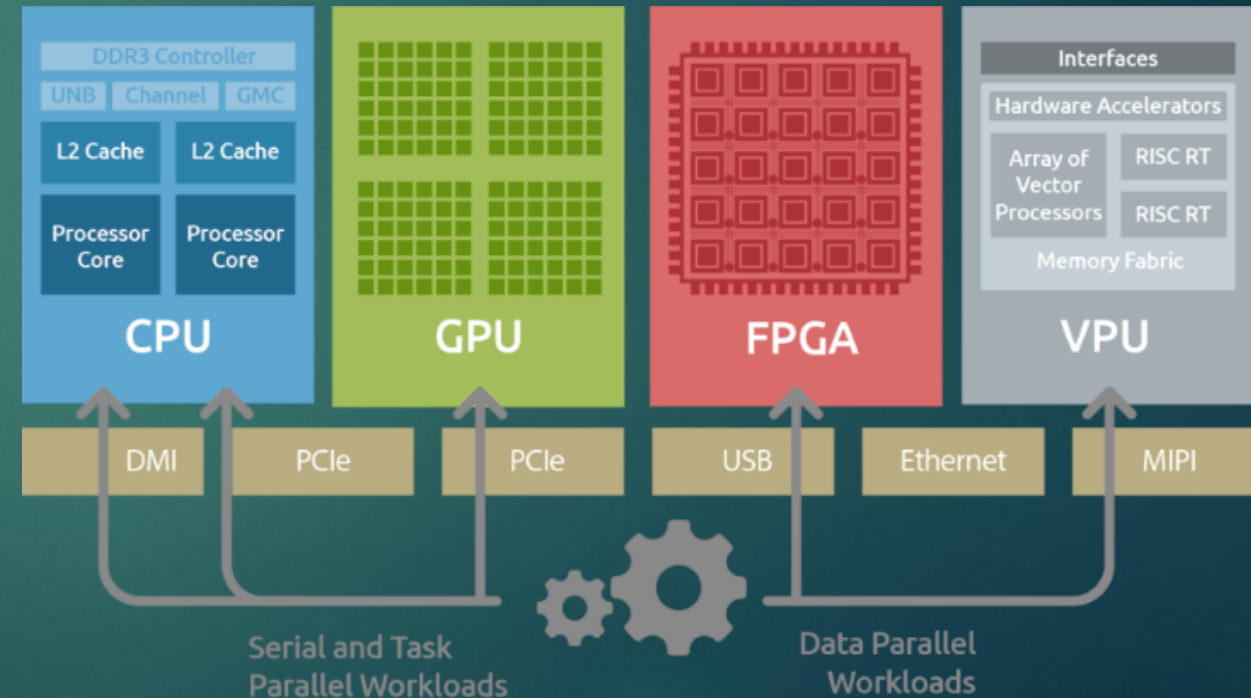
- ▶ Multiple Computer work together
- ▶ A form of Multiprocessor
- ▶ Each Computer has Processor and Memory
- ▶ Parallel Processing
- ▶ Many Forms
 - ▶ Cluster
 - ▶ Distributed
 - ▶ ...



Hardware Dedicated Accelerators

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- ▶ Dedicated Hardware for specific tasks, computing tasks
- ▶ Video / GPU Accelerator
- ▶ FPGA Accelerator
- ▶ ASIC Accelerator
- ▶ TPU Accelerator



Graphics Processing Unit (GPU) Evolution

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- ▶ Video Display Card/Unit
 - ▶ Graphic Display to Monitor
 - ▶ More Graphics Processing
 - ▶ Add Processing Capabilities
 - ▶ Graphics Processing Unit (GPU)
 - ▶ General Purposed GPU
- ▶ Massively Parallel Processing
 - ▶ Processing Elements
 - ▶ Internal Memory
- ▶ Host Control

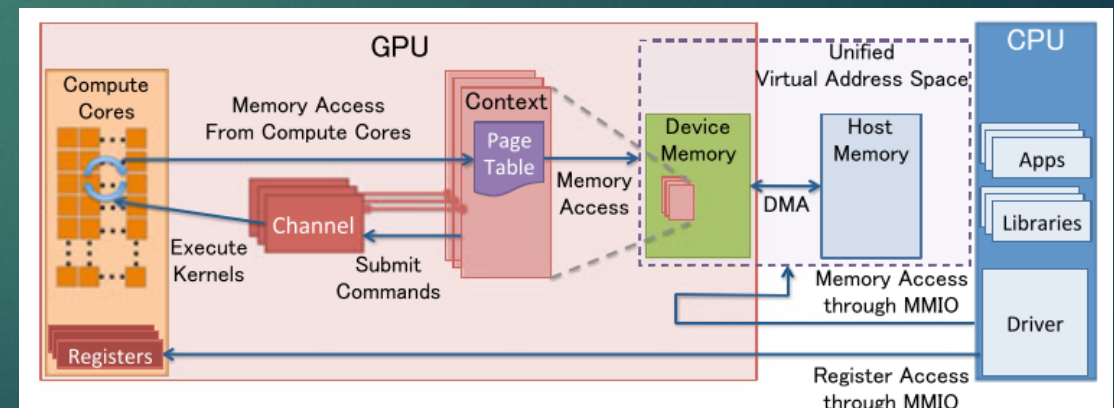
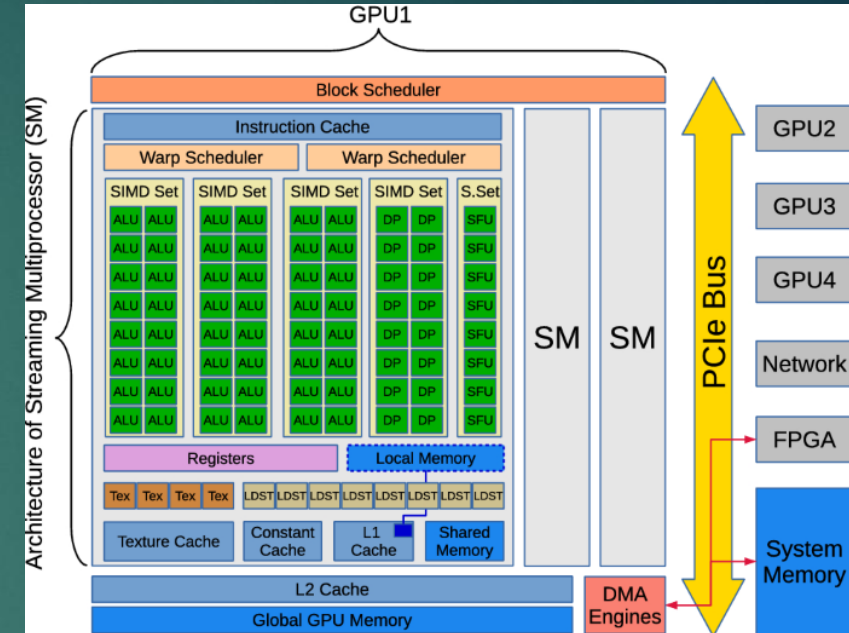
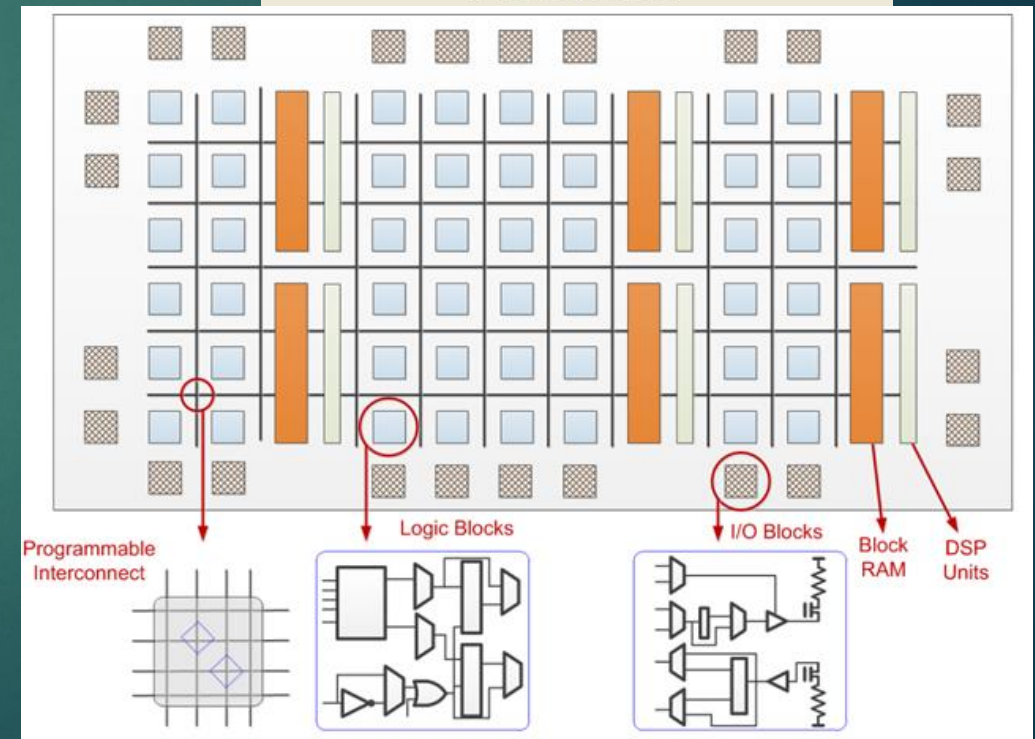
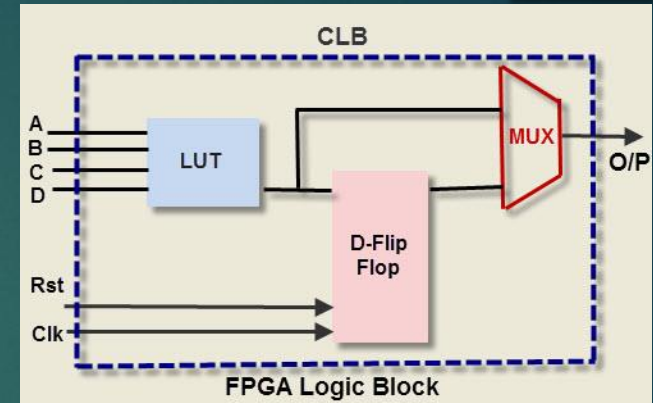


Fig. 1. GPU resource management model.

Field Programmable Gate Array (FPGA)

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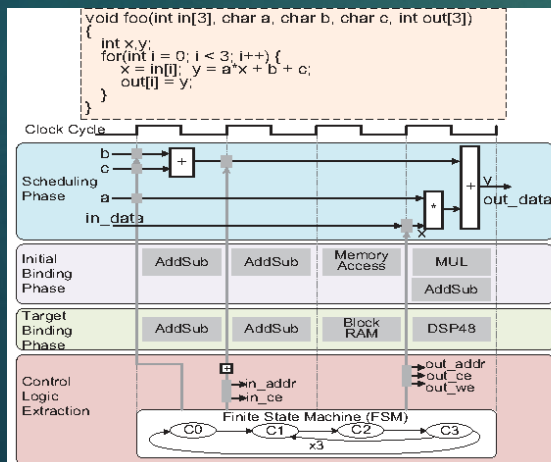
- ▶ Programmable Hardware / Digital Circuits
- ▶ Compose of
 - ▶ Configurable Logic Block (CLB)
 - ▶ Switch Block / Interconnection
 - ▶ IO Block
 - ▶ Hard Block (DSP, BRAM, CPU, HS, CLK, ...)
- ▶ Program by Hardware Description Language (HDL)
 - ▶ Verilog
 - ▶ VHDL (VHSIC (Very High-Speed Integrated Circuits) Hardware Description Language)
- ▶ Parallel Processing by Hardware / Digital Circuits



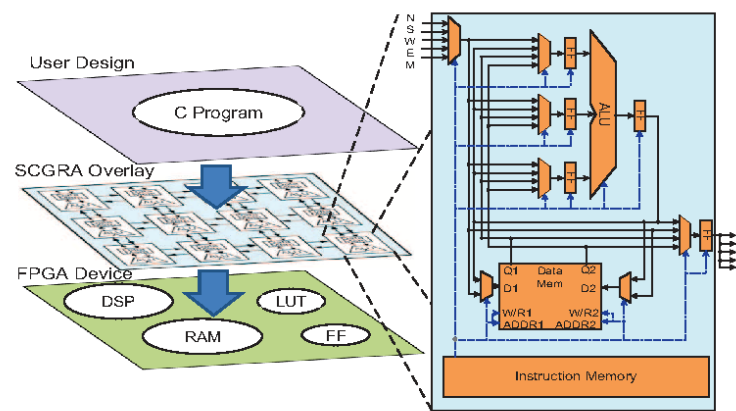
Field Programmable Gate Array (FPGA)

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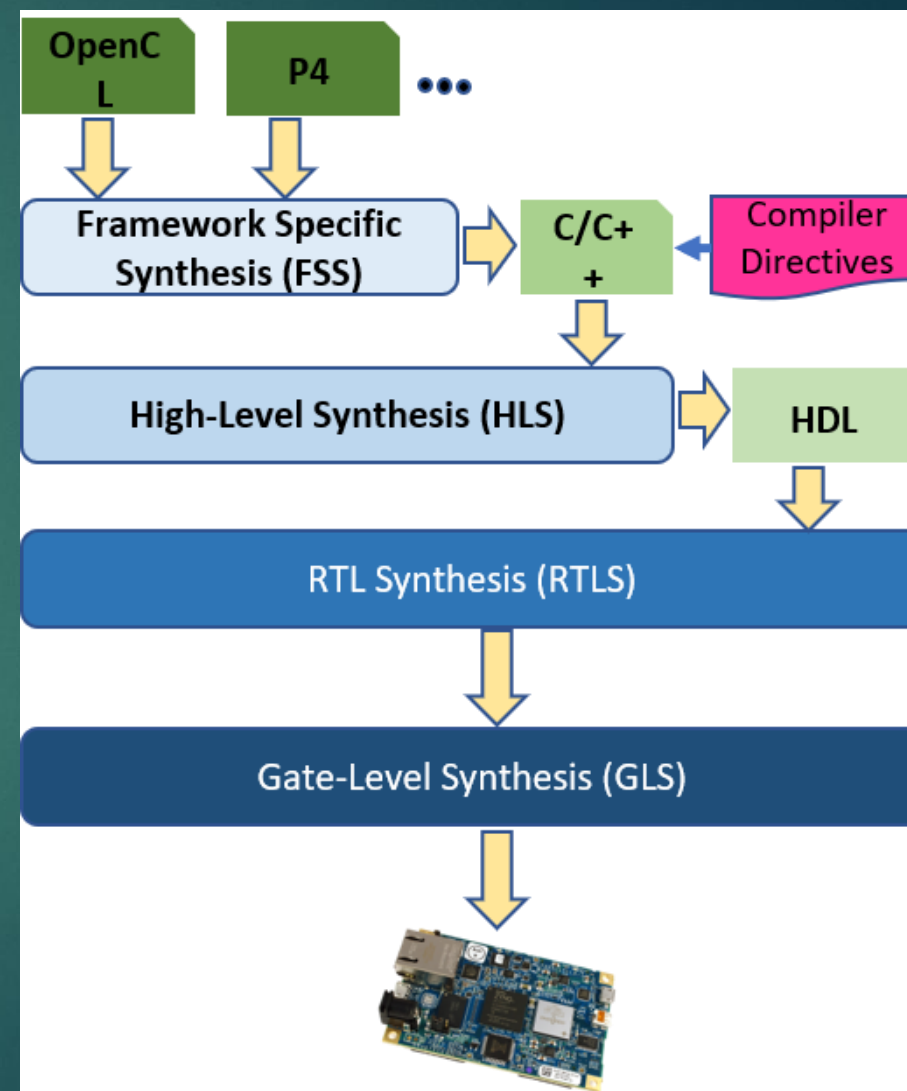
- ▶ High Level Synthesis uses for High Level Coding
 - ▶ Translate C/C++ (or Others) to HDL
- ▶ Compile HDL to Configuration Code
- ▶ Host Control by PCIe (or Others) Interfacing



(a) Scheduling, Binding, and Control Logic Extraction phases in Vivado HLS compilation flow



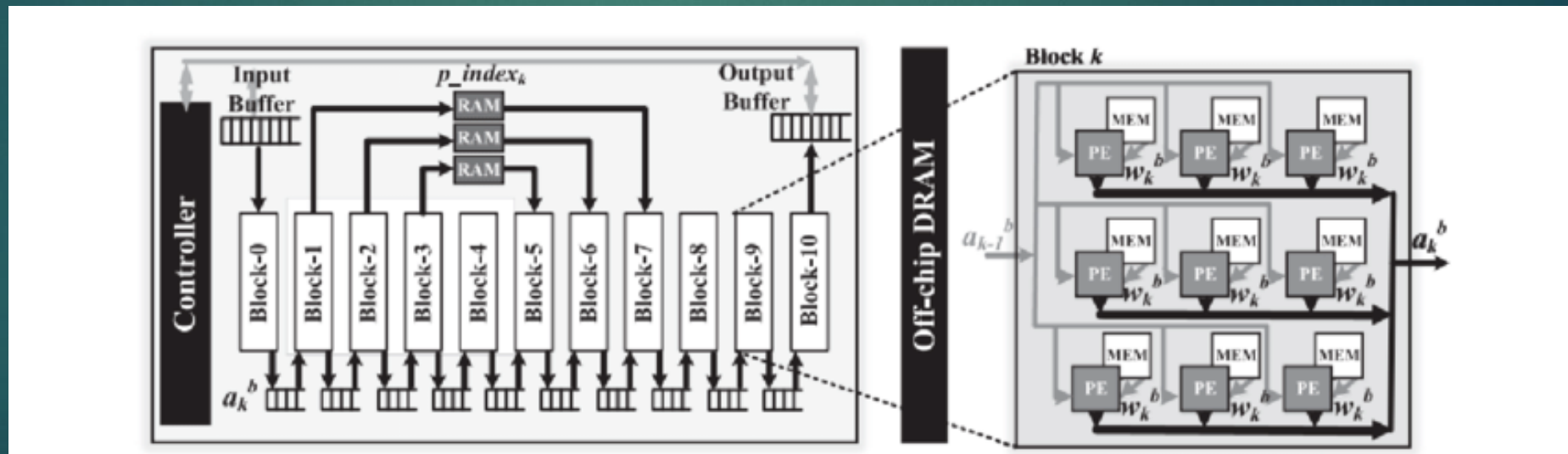
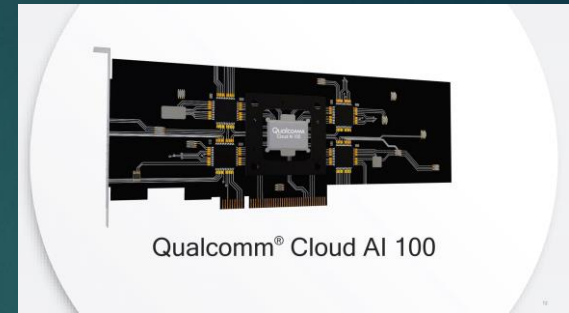
(b) ArchSyn using an SCGRA overlay between user design and FPGA device



Application Specific Integrated Circuit (ASIC)

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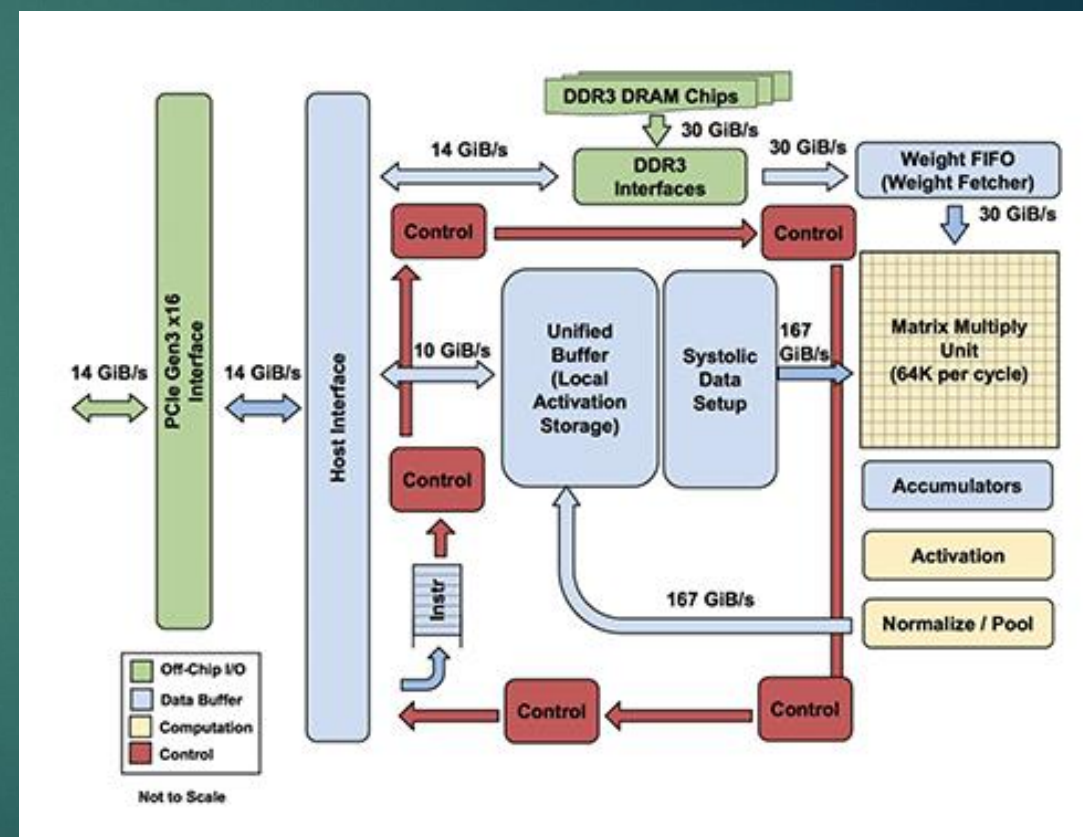
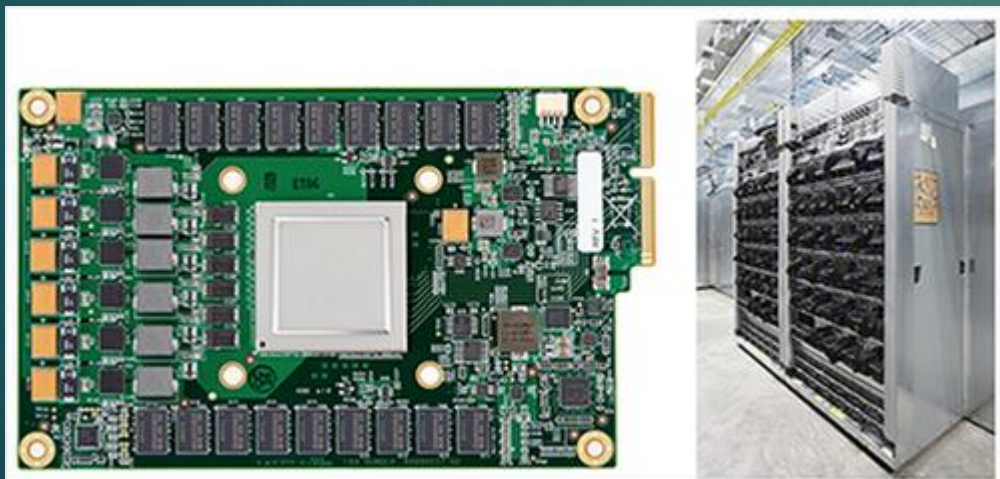
- ▶ Special Designed Hardware for Specific Tasks
- ▶ Optimized for Specific Processing



Tensor Processing Unit (TPU)

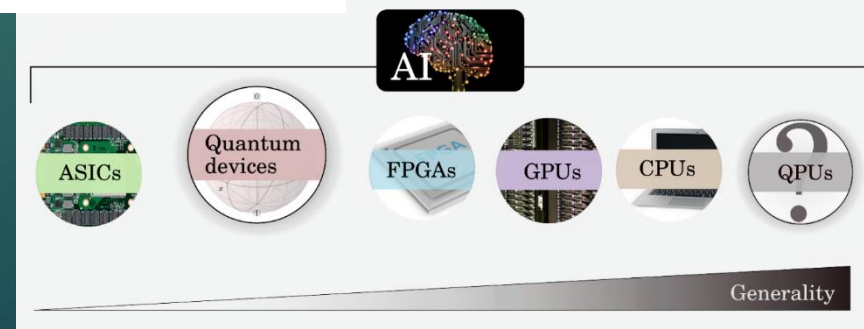
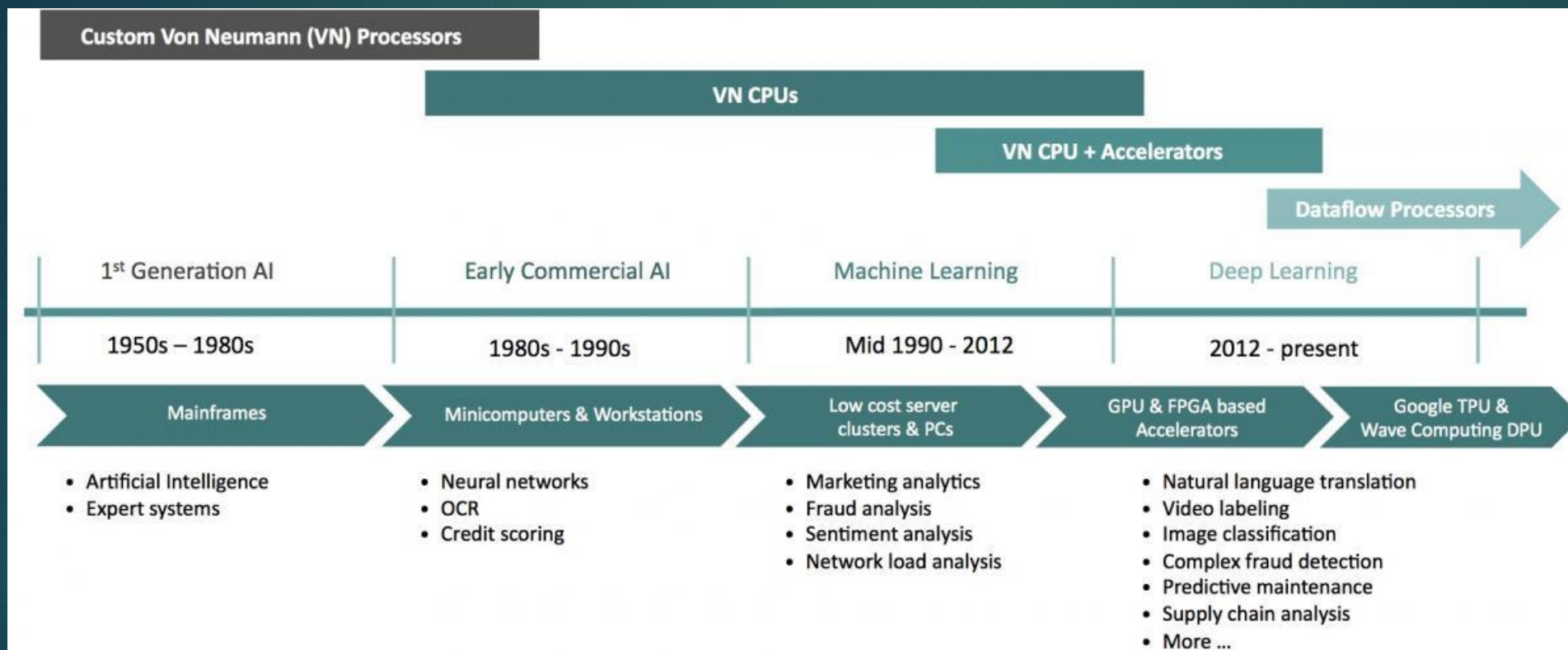
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- ▶ ASIC Accelerator by Google
- ▶ Parallel Number Processing



Computing Technology Trend

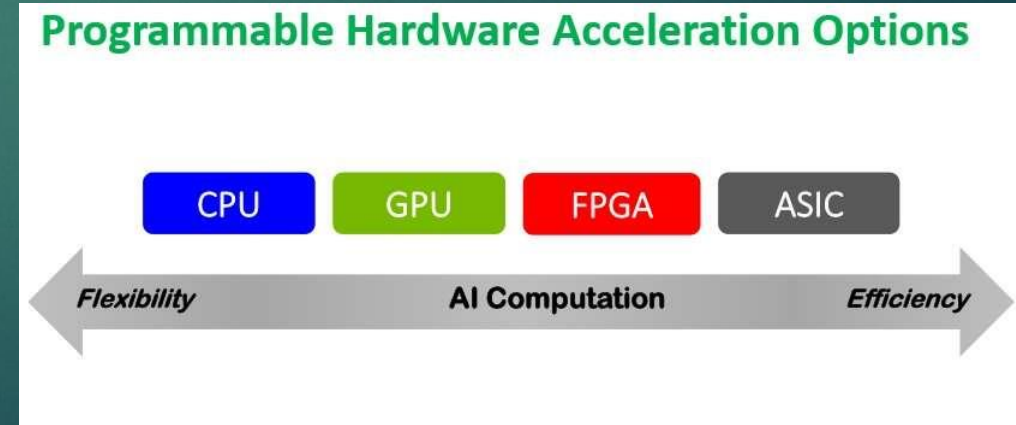
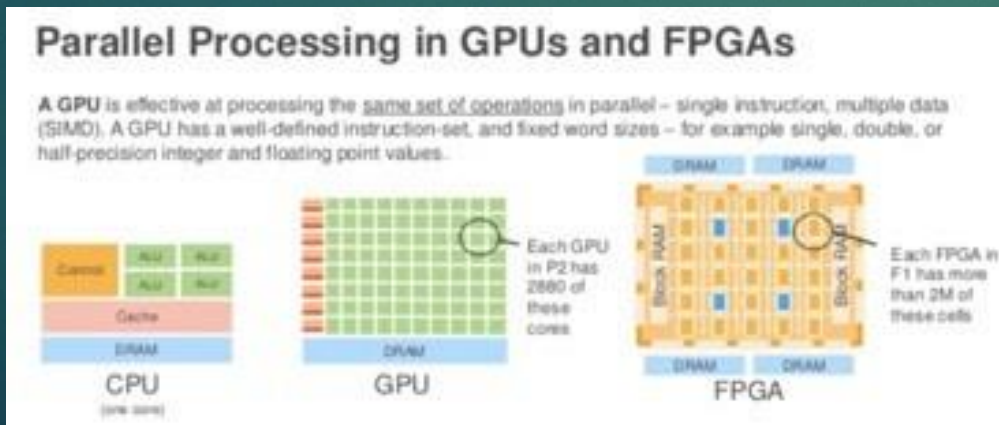
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Conclusion

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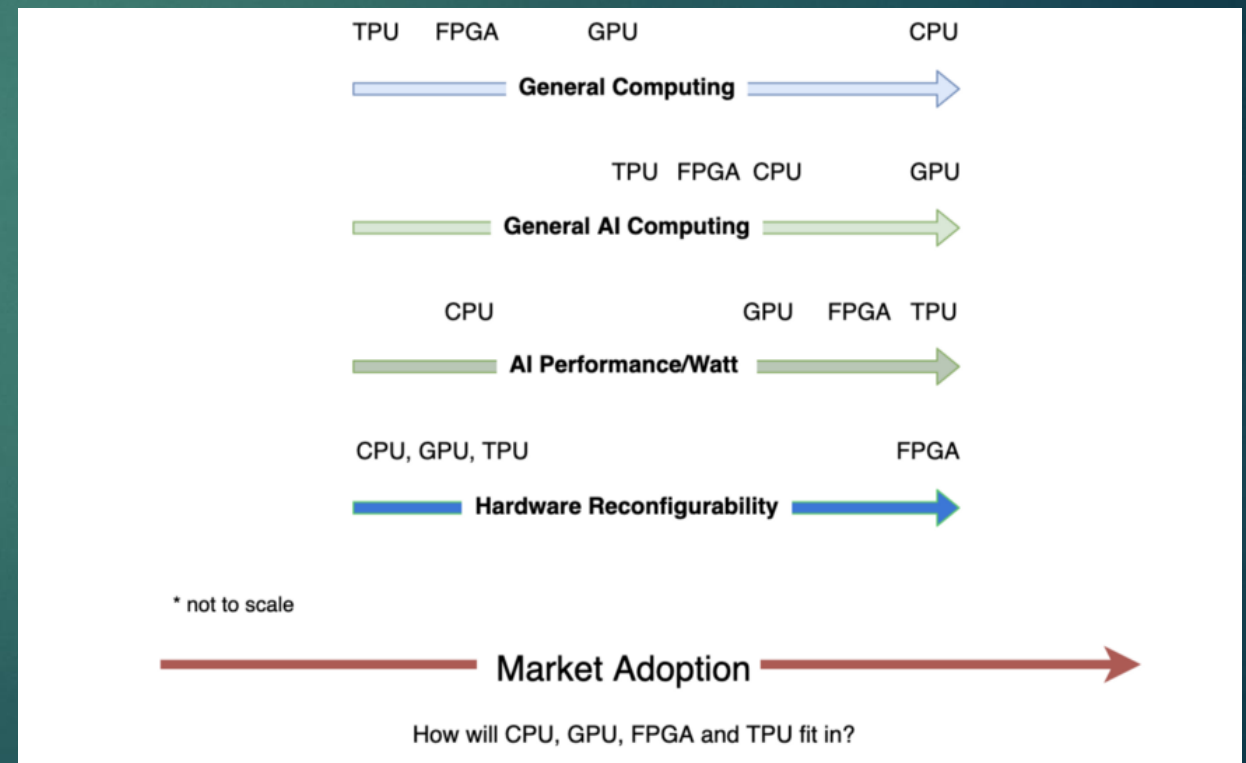
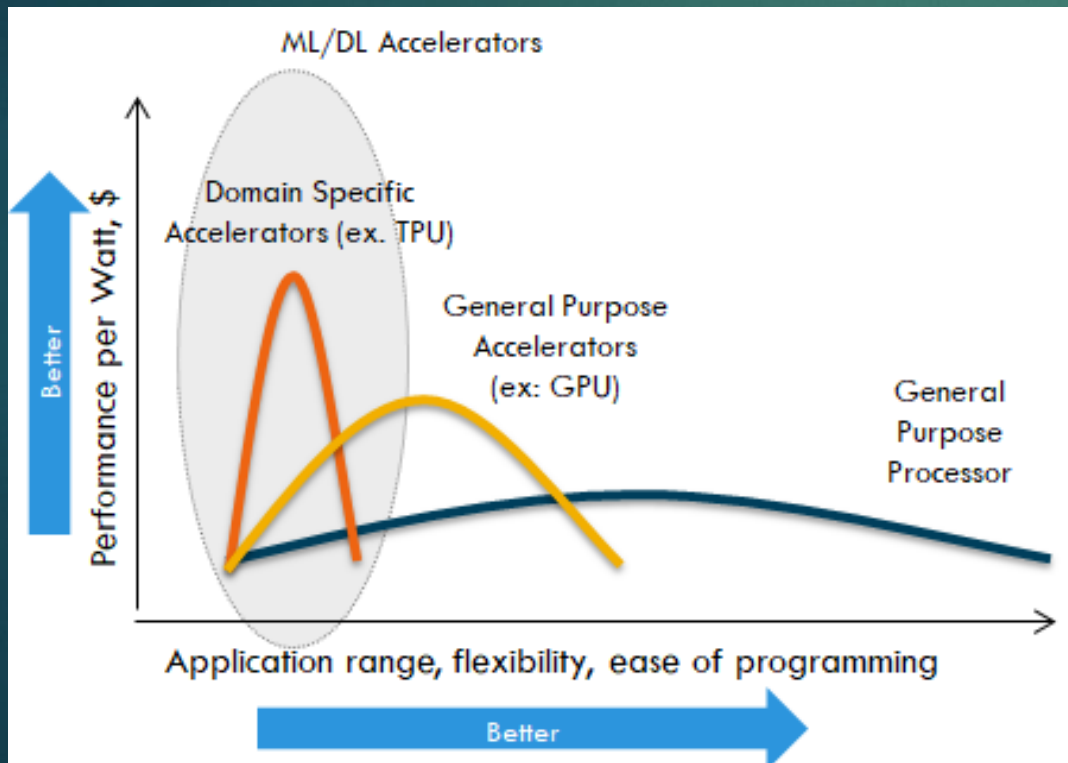
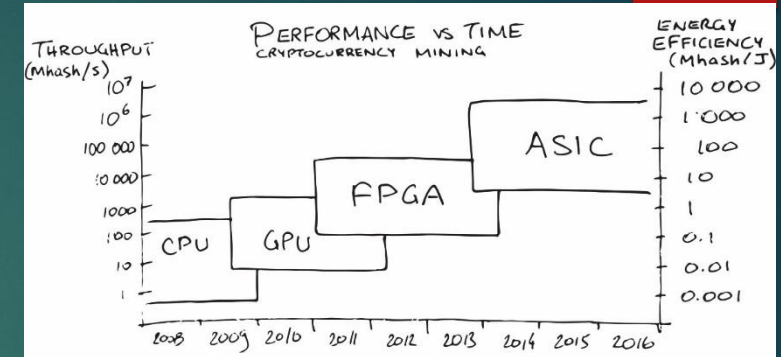
- ▶ Increasing data for processing drive Computing Technology
- ▶ Computing Technology improved for Speed/Performance
- ▶ Increasing Dedicated Parallel Hardware Resources => Accelerators
 - ▶ Processor-Based => Ready-Designed Hardware with Code Executing
 - ▶ Single CPU => Multiple CPU => Massive Processors => GPU
 - ▶ Hardware-Based => Digital Circuits for Executing
 - ▶ FPGA => ASIC (TPU)



Conclusion

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- ▶ Computing Accelerators make more parallel processing



END

QUESTIONS & ANSWERS